

Program : Diploma in Instrumentation Engineering	
Course Code : 3082	Course Title: Sensors and Transducers
Semester : 3	Credits: 3
Course Category: Program Core	
Periods per week: 3 (L:2 T:1 P:0)	Periods per semester: 45

Course Objectives:

- To introduce the basic principles of different types of sensors and transducers
- To learn different transducers and their basic applications.
- To know the basic idea about smart sensors, smart transmitters and industrial transmitters.

Course Prerequisites:

Topic	Course code	Course name	Semester
Basic principle of resistance, inductance, capacitance, diode, transistors etc.		Fundamentals of Electrical & and electronics	2
Permeability, Electromagnetic induction, self and mutual inductance; Transformer.		Fundamentals of Electrical & and electronics.	2

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Identify different types of resistive transducers.	13	Applying
CO2	Explain different types of inductive, capacitive and piezo electric transducers.	10	Understanding
CO3	Explain different types of photo electric transducers, radiation detectors, ultrasonic detectors, Hall effect transducers.	10	Understanding

CO4	Explain the concepts of transmitters, smart sensors, smart transmitters and basic idea about industrial transmitters.	10	Understanding
	Series Test	2	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	3					
CO2	3						
CO3	3						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Identify different types of resistive transducers.		
M1.01	Explain transducers and sensors and classify them.	5	Understanding
M1.02	Identify different types of resistive transducers and loading effect.	5	Applying
M1.03	Explain the applications of different types of resistive transducers.	3	Understanding
Contents: Introduction to Transducers, Resistive transducers: Define transducer and Sensor; classification of transducers as active-passive, analog-digital, unidirectional - bidirectional, mechanical – electrical, primary - secondary transducers; Resistive Transducers: Basic principle of resistive transducers; Working of linear potentiometer, loading effect; Basic principle of strain gauge, different types, derivation of equation for gauge factor, Construction and working of Light dependant resistor (LDR), Resistance temperature detector (RTD) and Thermistor; Displacement measurement using potentiometer; Weight measurement using strain gauge;			
CO2	Explain different types of inductive, capacitive and piezo electric transducers.		
M2.01	Explain the basic principle of inductive transducers and classification of inductive transducers.	2	Understanding

M2.02	Explain the construction and working of LVDT and eddy current transducer and LVDT for pressure measurement.	3	Understanding
M2.03	Explain basic principle of capacitive transducers and working of different types of capacitive transducers and capacitive level gauge.	3	Understanding
M2.04	Explain working principle of piezo electric transducers and piezo electric pressure transducer.	2	Understanding
	Series Test – I	1	

Contents:

Inductive, Capacitive and Piezo electric transducers: Basic principle of inductive transducer; Classification of inductive transducers; Construction and working of Linear Variable Differential Transformer (LVDT); pressure measurement using LVDT ; basic principle of Eddy current transducers; Illustrate basic principle of capacitive transducer; Working of different types of capacitive transducers (variable area, variable distance, variable dielectric type); Level measurement using capacitive transducer; Basic principle of piezo electric transducer; Equation for voltage sensitivity and charge sensitivity of the piezo electric transducer; Equivalent circuit of piezo electric transducer; Pressure measurement using piezo electric transducer.

CO3	Explain different types of photo electric transducers, radiation detectors, ultrasonic detectors, Hall effect transducers.		
M3.01	Explain photoelectric transducers- photo conductive (photo diode, photo transistor), photo emissive transducers (photo multiplier tube), solar cell and photo voltaic cell.	5	Understanding
M3.02	Explain the basic principle of Ultrasonic transducer and level measurement application.	2	Understanding
M3.03	Explain the construction and working principle of Hall effect transducer and current measurement application.	3	Understanding

Contents:

Photoelectric, Ultrasonic and Hall Effect Transducers: Illustrate photoelectric transducers-Working principle of photo conductive -photo diode, photo transistor- transducers. Working principle of photo emissive -photo multiplier tube- transducers, Working principle of solar cell and photo voltaic cell.

Working Principle of Ultra Sonic transducer. Construction and working principle of Hall effect transducer. Application of Ultrasonic transducer for level measurement, Hall Effect Transducers for current measurement.

CO4	Explain the concepts of transmitters, smart sensors, smart transmitters and basic idea about industrial transmitters.		
M4.01	Explain the concept of transmitters	2	Understanding
M4.02	Explain the Smart Sensors and Smart Transmitters using basic block diagram- Nano Sensors and pressure measurement using Nano Sensors.	6	Understanding
M4.03	Explain the basic concepts about industrial transmitters.	2	Understanding
	Series Test – II	1	
Contents: Transmitters, Smart Sensors, Smart Transmitters, Nano Sensors: Concept of standard electrical (4-20mA system) transmission; Smart sensors- basic block diagram. Smart Transmitters- basic block diagram: Nano sensor basics with example- Industrial transmitters-Direct mount and remote seal type transmitters- Capacitive type differential pressure transmitters.			

Text / Reference:

T/R	Book Title / Author
R1	A.K. Sawhney, “A course in Electrical and Electronic measurement and Instrumentation”, Dhanpathrai and Co.
R2	D.V.S. Murthy, “ Transducers and Instrumentation”, PHI. Pvt. Ltd.
R3	D.Patranabis, “ Principles of Industrial Instrumentation”, Tata McGraw-Hill. Education,
R4	J.B.Gupta, “Electrical and Electronic Measurements and instrumentation”, Kataria Pulishers.
R5	B. C. Nakra, K. K. Chaudhry, “Instrumentation, Measurement And Analysis”, Tata McGraw-Hill Publishers.
R6	Bela G. Liptak, “Process Measurement and Analysis”, Chilton Book Company.
R7	Ranganathan S, “ Transducer engineering ”, Allied publishers.
R8	R.K. JAIN, “Mechanical and Industrial Measurements”, Khanna Publishers.
R9	Krishna Kant, “Computer-Based Industrial Control”, PHI.
R10	D. Patranabis, “ Sensors and Transducers”, PHI.
R11	Ernest O. Doebelin, “Doebelin's Measurement Systems”, McGraw-Hill.

Online Resources:

Sl.No	Website Link
1	https://www.youtube.com/watch?v=FYggY7LLDrw
2	https://www.slideshare.net/manash234/classification-of-transducers
3	https://www.electronics-tutorials.ws/io/io_1.html
4	https://www.egr.msu.edu/classes/ece445/mason/Files/4-Sensors_ch2.pdf
5	https://www.youtube.com/watch?v=XSbUnUADlno
6	http://www.kyowa-ei.com/eng/technical/strain_gages/principles.html
7	https://www.youtube.com/watch?v=anCnrtjNLQM
8	http://www.te.com/content/dam/te-com/documents/sensors/global/lvdt-tutorial.pdf
9	https://www.electronics-tutorials.ws/electromagnetism/hall-effect.html
10	https://www.brighthubengineering.com/hvac/53147-how-capacitive-transducers-work/
11	https://www.powershow.com/view4/556e94-ODlIO/TRANSDUCERS_VARIABLE_RESISTIVE_CAPACITIVE_INDUCTIVE_powerpoint_ppt_presentation
12	https://www.ctscorp.com/resource-center/tutorials/piezo-basics/
13	https://www.youtube.com/watch?v=CckMERCuXvE
14	https://www.youtube.com/watch?v=7jQ12uUqK6E